Math GU4053 Intro to Algebraic Topology
Homework 5

due: Wednesday, March 4, 2026, before 23:59pm

submit the completed assignment to https://www.gradescope.com

- This assignment has **3 questions** on **two pages**.
- This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
- Your solutions must be *handwritten*. For instance on a tablet/iPad, or with pen and paper scanned to a pdf.
- While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
- Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
- Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)

1. Recall that a *based space* is a pair (X, x) with X a topological space and $x \in X$. If (X, x) and (Y, y) are based spaces, let $C((X, x), (Y, y))$ be the set of pairs $(f, [\lambda])$, where $f : X \to Y$ is a continuous map, $\lambda : I \to Y$ is a path with $\lambda(0) = y$ and $\lambda(1) = f(x)$, and $[\lambda]$ denotes its homotopy class relative to ∂I.

If (Z, z) is a third based space and $(f, [\lambda]) \in C((X, x), (Y, y))$ and $(g, [\mu]) \in C((Y, y), (Z, z))$, define $(g, [\mu]) \circ_C (f, [\lambda]) = (g \circ f, [\rho])$ where $\rho : I \to Z$ is the concatenation of the paths $g \circ \lambda : I \to Z$ and μ.

(a) Briefly explain why $(g, [\mu]) \circ_C (f, [\lambda])$ is a (well defined) element of $C((X, x), (Z, z))$.

(b) Briefly explain why $\circ_C$ is associative, define identity elements $1_{(X,x)} \in C((X, x), (X, x))$, and verify that C is in fact a category.

(c) Let $\phi = (f, [\lambda]) \in C((X, x), (Y, y))$ be a morphism, and assume that $f : X \to Y$ is a homeomorphism. Prove that ϕ is an isomorphism in the category C. Conversely, prove that if ϕ is an isomorphism, then f is a homeomorphism.

Remark (not needed for doing the problem): To a morphism $(f, [\mu]) : (X, x) \to (Y, y)$ in this category we can assign

$$\begin{aligned} \pi_1(X, x) &\to \pi_1(Y, y) \\ [\lambda] &\mapsto [\mu * (f \circ \lambda) * \overline{\mu}], \end{aligned}$$

which is easily seen to be a (well-defined) group homomorphism.
In this way the fundamental group can be made into a functor from C to the category of groups, simultaneously encoding both the change-of-basepoint isomorphisms and the usual functoriality with respect to based maps.

2. For a topological space X, let $\pi_0(X)$ be the set of path components in X. In other words, $\pi_0(X) = X/\sim$, where $\sim$ is the equivalence relation on X defined by $x \sim x'$ if there exists a continuous map $\lambda : I \to X$ with $\lambda(0) = x$ and $\lambda(1) = x'$.

Explain how to define for each continuous map $f : X \to Y$ a function $\pi_0(f) : \pi_0(X) \to \pi_0(Y)$, in a way that makes π_0 into a functor from topological spaces to sets.

3. For an abelian group A and an integer $n \in \mathbb{Z}$, we say that an element $a \in A$ is *divisible* by n if there exists some $a' \in A$ with $a = na'$.

Let $\mathbb{Z}S$ be the free abelian group generated by some set S. In other words, the set of formal linear combinations

$$c = \sum_{s \in S} c_s.s$$

with $c_s \in \mathbb{Z}$ for all $s \in S$, and such that $c_s = 0$ for all but finitely many $s \in S$.

(a) Let $c \in \mathbb{Z}S$ and assume $c \neq 0$. Prove that there are only finitely many integers $n \in \mathbb{Z}$ such that c is divisible by n.

(b) Let $\mathbb{Q}$ denote the rational numbers, regarded as an abelian group with respect to addition. Prove that $1 \in \mathbb{Q}$ is divisible by any $n \in \mathbb{Z} \setminus \{0\}$.

(c) Deduce that there does not exist a set S such that $\mathbb{Q}$ is isomorphic to $\mathbb{Z}S$.